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Solenoid valve manifold

Abstract

An electromagnetic valve (11) includes an electromagnetic valve-side connector unit (55, 65). A base (30) includes a base-side connector unit (80, 90) to which the electromagnetic valve-side connector unit (55, 65) is connected. The base (30) includes an insertion port (43, 44) into which the electromagnetic valve-side connector unit (55, 65) is inserted. The base-side connector unit (80, 90) is disposed inside the insertion port (43, 44). The electromagnetic valve (11) includes a protrusion (56, 66). The base (30) includes a recess (87, 97) configured to guide the protrusion (56, 66). When the protrusion (56, 66) is guided by the recess (87, 97), the electromagnetic valve-side connector unit (55, 65) is inserted into the insertion port (43, 44) without interfering with a section of the base (30) around the insertion port (43, 44) and is connected to the base-side connector unit (80, 90).

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0003610	12/2018	Lucidera	N/A	F16K 11/0716
2020/0149650	12/2019	Kidder	N/A	F16K 27/048
2021/0131459	12/2020	Weickel	N/A	F16K 37/005
2021/0324966	12/2020	Hayashi et al.	N/A	N/A
2023/0137993	12/2022	Ohki	251/129.15	F15B 13/0857

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H09329269	12/1996	JP	N/A
2020186781	12/2019	JP	N/A

OTHER PUBLICATIONS

International Search Report (English and Japanese) and Written Opinion for International Patent Application No. PCT/JP2023/011352, dated May 30, 2023, 8 pages. cited by applicant

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATIONS**

(1) This application is a U.S. National Stage Application filed under 35 U.S.C. § 371 of PCT/JP2023/011352, filed Mar. 23, 2023, and entitled “SOLENOID VALVE MANIFOLD”, which

claims priority from Japanese Patent Application No. 2022-064676, filed on Apr. 8, 2022. The entire contents of each of the above-identified patent applications are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an electromagnetic valve manifold.

BACKGROUND ART

(3) For example, Patent Literature 1 discloses an electromagnetic valve manifold that includes an electromagnetic valve, a control unit, and a base. The electromagnetic valve includes a solenoid unit. The electromagnetic valve includes an electromagnetic valve-side connector unit. The electromagnetic valve-side connector unit is electrically connected to the solenoid unit. The control unit includes a circuit board. The circuit board controls operation of the electromagnetic valve. The base includes a control unit. The base includes a base-side connector unit. The electromagnetic valve-side connector unit is connected to the base-side connector unit. The base-side connector unit electrically connects the electromagnetic valve-side connector unit and the circuit board to each other. The base includes an insertion port. The base-side connector unit is disposed inside the insertion port. The electromagnetic valve-side connector unit is inserted into the insertion port. When the electromagnetic valve-side connector unit inserted into the insertion port is connected to the base-side connector unit, the electromagnetic valve-side connector unit and the circuit board are electrically connected to each other via the base-side connector unit.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Laid-Open Patent Publication No. 2020-186781

SUMMARY OF INVENTION

Technical Problem

(5) In such an electromagnetic valve manifold, when the electromagnetic valve-side connector unit is connected to the base-side connector unit, the electromagnetic valve-side connector unit may interfere with a section of the base around the insertion port. Therefore, there is a desire to improve the ease of mounting for electromagnetic valve manifolds.

Solution to Problem

(6) In one general aspect, an electromagnetic valve manifold includes an electromagnetic valve including a solenoid unit, a control unit including a circuit board configured to control operation of the electromagnetic valve, and a base that includes the control unit. The electromagnetic valve includes an electromagnetic valve-side connector unit electrically connected to the solenoid unit. The base includes a base-side connector unit to which the electromagnetic valve-side connector unit is connected. The base-side connector unit is configured to electrically connect the electromagnetic valve-side connector unit and the circuit board to each other. The base includes an insertion port into which the electromagnetic valve-side connector unit is inserted. The base-side connector unit is disposed inside the insertion port. One of the electromagnetic valve or the base includes a protrusion. The other one of the electromagnetic valve or the base includes a recess. The recess is configured to guide the protrusion in a same direction as an insertion direction of the electromagnetic valve-side connector unit into the insertion port. When the protrusion is guided by the recess, the electromagnetic valve-side connector unit is inserted into the insertion port without interfering with a section of the base around the insertion port and is connected to the base-side connector unit.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a cross-sectional view of an electromagnetic valve manifold according to one

embodiment.

(2) FIG. 2 is an exploded perspective view of an electromagnetic valve and a base of FIG. 1.

(3) FIG. 3 is an enlarged partial cross-sectional view of the electromagnetic valve manifold shown in FIG. 1.

(4) FIG. 4 is a cross-sectional view illustrating a state in which the connector member shown in FIG. 3 is attached to the base.

(5) FIG. 5 is an enlarged exploded perspective view showing part of the electromagnetic valve and the base shown in FIG. 2.

(6) FIG. 6 is a plan view showing a part of the base shown in FIG. 5.

(7) FIG. 7 is an enlarged cross-sectional view showing part of the electromagnetic valve and the base shown in FIG. 5.

(8) FIG. 8 is a cross-sectional view showing a state before the electromagnetic valve shown in FIG. 3 is mounted to the base.

(9) FIG. 9 is a cross-sectional view showing a state before the electromagnetic valve shown in FIG. 3 is mounted to the base.

(10) FIG. 10 is an enlarged cross-sectional view showing part of an electromagnetic valve and a base according to another embodiment.

(11) FIG. 11 is an enlarged cross-sectional view showing part of an electromagnetic valve and a base according to another embodiment.

DESCRIPTION OF EMBODIMENTS

(12) An electromagnetic valve manifold according to one embodiment will now be described with reference to FIGS. 1 to 9.

(13) Basic Configuration of Electromagnetic Valve Manifold

(14) As shown in FIG. 1, an electromagnetic valve manifold **10** includes an electromagnetic valve **11**. The electromagnetic valve **11** includes a main valve unit **12**, a first pilot electromagnetic valve **50**, and a second pilot electromagnetic valve **60**. The electromagnetic valve **11** is a double-solenoid electromagnetic pilot valve. The electromagnetic valve manifold **10** includes a control unit **40** and a base **30**.

(15) The main valve unit **12** includes a valve casing **13**. The valve casing **13** has the shape of an elongated rectangular block. The valve casing **13** includes a valve body **14**, a first coupling block **15**, and a second coupling block **16**. The valve body **14** has the shape of an elongated rectangular block. The first coupling block **15** is coupled to a first end in the longitudinal direction of the valve body **14**. The second coupling block **16** is coupled to a second end in the longitudinal direction of the valve body **14**. The valve body **14** includes a base facing surface **14a**, which faces the base **30**.

(16) The valve casing **13** includes a spool valve hole **17**. The spool valve hole **17** is formed in the valve body **14**. The spool valve hole **17** is a circular hole. The spool valve hole **17** extends in the longitudinal direction of the valve body **14**. A first end of the spool valve hole **17** opens in a first end face in the longitudinal direction of the valve body **14**. A second end of the spool valve hole **17** opens in a second end face in the longitudinal direction of the valve body **14**. Thus, the spool valve hole **17** extends through the valve body **14** in the longitudinal direction.

(17) The electromagnetic valve **11** includes a spool valve **18**. The spool valve **18** is accommodated in the spool valve hole **17**. The spool valve **18** is accommodated in the spool valve hole **17** with the axial direction of the spool valve **18** agreeing with the axial direction of the spool valve hole **17**. The spool valve **18** is reciprocally accommodated in the spool valve hole **17**.

(18) The electromagnetic valve **11** includes a supply port P, a first output port A, a second output port B, a first discharge port R1, and a second discharge port R2. The electromagnetic valve **11** of the present embodiment is thus a five-port electromagnetic valve. The supply port P, the first output port A, the second output port B, the first discharge port R1, and the second discharge port R2 are formed in the valve body **14**. The supply port P, the first output port A, the second output port B, the first discharge port R1, and the second discharge port R2 are each connected to the spool valve

hole **17**.

(19) The first discharge port **R1**, the first output port **A**, the supply port **P**, the second output port **B**, and the second discharge port **R2** are arranged in that order from the first end to the second end in the longitudinal direction of the valve body **14**. First ends of the supply port **P**, the first output port **A**, the second output port **B**, the first discharge port **R1**, and the second discharge port **R2** are each connected to the spool valve hole **17**. Second ends of the supply port **P**, the first output port **A**, the second output port **B**, the first discharge port **R1**, and the second discharge port **R2** each open in the base facing surface **14a** of the valve body **14**.

(20) The electromagnetic valve **11** includes a first piston **19** and a second piston **20**. The first piston **19** has the shape of a disc. The first piston **19** is coupled to a first end of the spool valve **18**. The first piston **19** moves integrally with the spool valve **18**. The second piston **20** has the shape of a disc. The second piston **20** is coupled to a second end of the spool valve **18**. The second piston **20** moves integrally with the spool valve **18**.

(21) The first coupling block **15** includes a first piston accommodating recess **21**, which is a circular hole. The first piston accommodating recess **21** accommodates the first piston **19**, while allowing the first piston **19** to reciprocate. The first piston accommodating recess **21** and the first piston **19** define a first pilot pressure chamber **22**. Pilot fluid is supplied to and discharged from the first pilot pressure chamber **22**.

(22) The second coupling block **16** includes a second piston accommodating recess **23**, which is a circular hole. The second piston accommodating recess **23** accommodates the second piston **20**, while allowing the second piston **20** to reciprocate. The second piston accommodating recess **23** and the second piston **20** define a second pilot pressure chamber **24**. Pilot fluid is supplied to and discharged from the second pilot pressure chamber **24**.

(23) The base **30** includes a manifold block **31**. The manifold block **31** has the shape of an elongated rectangular block. The manifold block **31** includes a placement surface **31a**. The electromagnetic valve **11** is mounted on the placement surface **31a**. The longitudinal direction of the manifold block **31** agrees with the longitudinal direction of the valve casing **13**.

(24) The manifold block **31** includes a supply passage **32**, a first output passage **33**, a second output passage **34**, a first discharge passage **35**, and a second discharge passage **36**. The supply passage **32**, the first output passage **33**, the second output passage **34**, the first discharge passage **35**, and the second discharge passage **36** each include a first end portion that opens in the placement surface **31a** and a second end portion opposite to the first end portion.

(25) The first end portion of the supply passage **32** is connected to the supply port **P**. The first end portion of the first output passage **33** is connected to the first output port **A**. The first end portion of the second output passage **34** is connected to the second output port **B**. The first end portion of the first discharge passage **35** is connected to the first discharge port **R1**. The first end portion of the second discharge passage **36** is connected to the second discharge port **R2**.

(26) The second end portion of the supply passage **32** is connected to a fluid supply source (not shown) via piping and the like. The second end portion of the first output passage **33** and the second end portion of the second output passage **34** are respectively connected to a fluid pressure device (not shown), for example, via piping and the like. The second end portion of the first discharge passage **35** and the second end portion of the second discharge passage **36** are connected to the atmosphere, for example, via piping and the like.

(27) The electromagnetic valve manifold **10** includes an annular gasket **37**. The gasket **37**, for example, has the shape of a thin plate. The gasket **37** is disposed between the valve body **14** of the electromagnetic valve **11** and the manifold block **31**. Thus, the gasket **37** is disposed between the electromagnetic valve **11** and the base **30**. The gasket **37** provides a seal between the valve body **14** of the electromagnetic valve **11** and the manifold block **31**.

(28) As shown in FIG. 2, the electromagnetic valve **11** is fixed to the base **30** with bolts **38**. The valve casing **13** of the electromagnetic valve **11** includes two bolt insertion holes **38h**, through

which the bolts **38** are inserted. For illustrative purposes, only one of the bolt insertion holes **38h** is shown in FIG. 2.

(29) The base **30** includes internal thread holes **39**. The bolts **38** are respectively threaded into the internal thread holes **39**. Two internal thread holes **39** are formed in the placement surface **31a** of the manifold block **31**. The internal thread holes **39** are formed near the gasket **37** in the placement surface **31a** of the manifold block **31**. Each bolt **38** inserted into the corresponding bolt insertion hole **38h** is fixed to the base **30** by being screwed into the corresponding internal thread hole **39** in a state in which the electromagnetic valve **11** is placed on the placement surface **31a**.

(30) As shown in FIG. 1, the control unit **40** is incorporated in the manifold block **31**. The base **30** thus includes the control unit **40**. The control unit **40** includes a circuit board **41**. The circuit board **41** receives power from an external controller such as a programmable logic controller (PLC: not shown). The circuit board **41** is incorporated in the manifold block **31**. The circuit board **41** controls operation of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**. The circuit board **41** thus controls operation of the electromagnetic valve **11**.

(31) The electromagnetic valve **11** includes a first solenoid unit **51** and a second solenoid unit **61**, which are solenoid units. The first pilot electromagnetic valve **50** includes the first solenoid unit **51**. The first pilot electromagnetic valve **50** supplies pilot fluid to and discharges pilot fluid from the first pilot pressure chamber **22**. When voltage is applied to the first solenoid unit **51** from the circuit board **41**, the first pilot electromagnetic valve **50** supplies compressed fluid from the fluid supply source (not shown), which is pilot fluid, to the first pilot pressure chamber **22**. When the application of voltage to the first solenoid unit **51** from the circuit board **41** is stopped, the first pilot electromagnetic valve **50** stops the supply of fluid from the fluid supply source to the first pilot pressure chamber **22**. The first pilot electromagnetic valve **50** discharges the pilot fluid in the first pilot pressure chamber **22** to the atmosphere.

(32) The second pilot electromagnetic valve **60** includes a second solenoid unit **61**. The second pilot electromagnetic valve **60** supplies pilot fluid to and discharges pilot fluid from the second pilot pressure chamber **24**. When voltage is applied to the second solenoid unit **61** from the circuit board **41**, the second pilot electromagnetic valve **60** supplies compressed fluid from the fluid supply source, which is pilot fluid, to the second pilot pressure chamber **24**. When the application of voltage to the second solenoid unit **61** from the circuit board **41** is stopped, the second pilot electromagnetic valve **60** stops the supply of fluid from the fluid supply source to the second pilot pressure chamber **24**. The second pilot electromagnetic valve **60** discharges the pilot fluid in the second pilot pressure chamber **24** to the atmosphere.

(33) The spool valve **18** is switchable between a first position and a second position. For example, there may be a case in which voltage is applied to the first solenoid unit **51** from the circuit board **41**, and voltage is not applied to the second solenoid unit **61** from the circuit board **41**. In this case, the first pilot electromagnetic valve **50** supplies compressed fluid, which is pilot fluid, from the fluid supply source to the first pilot pressure chamber **22**. On the other hand, the second pilot electromagnetic valve **60** discharges the pilot fluid in the second pilot pressure chamber **24** to the atmosphere. Accordingly, the spool valve **18** moves toward the second piston accommodating recess **23**. As a result, the spool valve **18** is switched to the first position, in which the supply port P is connected to the first output port A, and the second output port B is connected to the second discharge port R2. Also, when the spool valve **18** is switched to the first position, the supply port P and the second output port B are disconnected from each other, and the first output port A and the first discharge port R1 are disconnected from each other. Accordingly, the fluid supplied to the supply port P flows to the fluid pressure device via the first output port A, and the first output passage **33**. Then, the pressurized fluid from the fluid pressure device flows toward the second discharge passage **36** via the second output passage **34**, the second output port B, and the second discharge port R2 to be discharged to the outside.

(34) Also, for example, there may be a case in which voltage is not applied to the first solenoid unit

51 from the circuit board 41, and voltage is applied to the second solenoid unit 61 from the circuit board 41. In this case, the second pilot electromagnetic valve 60 supplies compressed fluid, which is pilot fluid, from the fluid supply source to the second pilot pressure chamber 24. On the other hand, the first pilot electromagnetic valve 50 discharges the pilot fluid in the first pilot pressure chamber 22 to the atmosphere. Accordingly, the spool valve 18 moves toward the first piston accommodating recess 21. As a result, the spool valve 18 is switched to the second position, in which the supply port P is connected to the second output port B, and the first output port A is connected to the first discharge port R1. Also, when the spool valve 18 is switched to the second position, the supply port P and the first output port A are disconnected from each other, and the second output port B and the second discharge port R2 are disconnected from each other. Accordingly, the fluid supplied to the supply port P flows to the fluid pressure device via the second output port B, and the second output passage 34. Then, the pressurized fluid from the fluid pressure device flows toward the first discharge passage 35 via the first output passage 33, the first output port A, and the first discharge port R1 to be discharged to the outside.

(35) In this manner, the first pilot electromagnetic valve 50 supplies pilot fluid to and discharges pilot fluid from the first pilot pressure chamber 22, and the second pilot electromagnetic valve 60 supplies pilot fluid to and discharges pilot fluid from the second pilot pressure chamber 24.

Accordingly, the spool valve 18 reciprocates in the spool valve hole 17 between the first position and the second position. The connection state among the ports is switched as the spool valve 18 is switched between the first position and the second position. Thus, the main valve unit 12 supplies pilot fluid to and discharges pilot fluid from each of the first pilot pressure chamber 22 and the second pilot pressure chamber 24, thereby switching the connections among the ports. FIG. 1 shows a state in which the spool valve 18 is located at the second position.

(36) The first pilot electromagnetic valve 50 and the second pilot electromagnetic valve 60 are integrated and arranged side by side. Specifically, the first pilot electromagnetic valve 50 and the second pilot electromagnetic valve 60 are located on a side of the first coupling block 15 that is opposite to the valve body 14. The first pilot electromagnetic valve 50 and the second pilot electromagnetic valve 60 are arranged side by side with the first coupling block 15. The first pilot electromagnetic valve 50 and the second pilot electromagnetic valve 60 are integrated with the first coupling block 15. Therefore, the first pilot electromagnetic valve 50 and the second pilot electromagnetic valve 60 are integrated with the main valve unit 12.

(37) The electromagnetic valve 11 includes a projecting portion 25. The projecting portion 25 is a portion of the electromagnetic valve 11 that projects further toward the base 30 than the base facing surface 14a of the valve body 14. The projecting portion 25 includes a part of the first pilot electromagnetic valve 50 and a part of the second pilot electromagnetic valve 60.

(38) As shown in FIG. 2, the projecting portion 25 includes a first side surface 25a and a second side surface 25b. The first side surface 25a and the second side surface 25b are located at opposite ends of the projecting portion 25 in the arrangement direction of the first pilot electromagnetic valve 50 and the second pilot electromagnetic valve 60. The first side surface 25a is a side surface of the first pilot electromagnetic valve 50. The second side surface 25b is a side surface of the second pilot electromagnetic valve 60. The first side surface 25a and the second side surface 25b protrude further toward the base 30 than the base facing surface 14a.

(39) Electromagnetic Valve-Side Connector Unit

(40) As shown in FIG. 3, the electromagnetic valve 11 includes a first electromagnetic valve-side connector unit 55 and a second electromagnetic valve-side connector unit 65, which are electromagnetic valve-side connector units. The first pilot electromagnetic valve 50 includes the first electromagnetic valve-side connector unit 55. The first electromagnetic valve-side connector unit 55 includes first conductive members 52, which are conductive members, and a first tubular portion 53, which is a tubular portion. The first conductive members 52 are electrically connected to the first solenoid unit 51. The first pilot electromagnetic valve 50 includes two first conductive

members **52**. One of the two first conductive members **52** forms a positive electrode line, and the other one of the two first conductive members **52** forms a negative electrode line. Each first conductive member **52** is, for example, columnar.

(41) The first tubular portion **53** surrounds the two first conductive members **52**. Specifically, the first tubular portion **53** surrounds the distal end portions, which are the end portions of the two first conductive members **52** on the side opposite to the first solenoid unit **51**. The first tubular portion **53** is cylindrical. The distal end of each first conductive member **52** is located inside the first tubular portion **53**. The axial direction of the first tubular portion **53** is the same as the extending direction of the distal end portion of each first conductive member **52**. The first pilot electromagnetic valve **50** is arranged with respect to the base **30** such that the opening of the first tubular portion **53** faces the base **30**. Thus, the distal end portion of each first conductive member **52** extends toward the base **30**. The first conductive members **52** and the first tubular portion **53** project from the distal end face of the projecting portion **25**.

(42) The first tubular portion **53** includes a first attachment groove **53a** in the outer circumferential surface. The first attachment groove **53a** is annular. A first lip packing **54** is attached to the first attachment groove **53a**. The first lip packing **54** is made of rubber. The first lip packing **54** is circular and annular.

(43) The second pilot electromagnetic valve **60** includes a second electromagnetic valve-side connector unit **65**. The second electromagnetic valve-side connector unit **65** includes second conductive members **62**, which are conductive members, and a second tubular portion **63**, which is a tubular portion. The second conductive members **62** are electrically connected to the second solenoid unit **61**. The second pilot electromagnetic valve **60** includes two second conductive members **62**. One of the two second conductive members **62** forms a positive electrode line, and the other one of the two second conductive members **62** forms a negative electrode line. Each second conductive member **62** is, for example, columnar.

(44) The second tubular portion **63** surrounds the two second conductive members **62**. Specifically, the second tubular portion **63** surrounds the distal end portions, which are the end portions of the two second conductive members **62** on the side opposite to the second solenoid unit **61**. The second tubular portion **63** is cylindrical. The distal end of each second conductive member **62** is located inside the second tubular portion **63**. The axial direction of the second tubular portion **63** is the same as the extending direction of the distal end portion of each second conductive member **62**. The second pilot electromagnetic valve **60** is arranged with respect to the base **30** such that the opening of the second tubular portion **63** faces the base **30**. Thus, the distal end portion of each second conductive member **62** extends toward the base **30**. The second conductive members **62** and the second tubular portion **63** project from the distal end face of the projecting portion **25**.

(45) The axial direction of the second tubular portion **63** is the same as the axial direction of the first tubular portion **53**. The extending direction of the distal end portion of each second conductive member **62** is the same as the extending direction of the distal end portion of each first conductive member **52**. The first tubular portion **53** and the second tubular portion **63** are arranged side by side in the arrangement direction of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**. Therefore, the arrangement direction of the first electromagnetic valve-side connector unit **55** and the second electromagnetic valve-side connector unit **65** is the same as the arrangement direction of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**.

(46) The second tubular portion **63** includes a second attachment groove **63a** in the outer circumferential surface. The second attachment groove **63a** is annular. A second lip packing **64** is attached to the second attachment groove **63a**. The second lip packing **64** is made of rubber. The second lip packing **64** is circular and annular.

(47) Base-Side Connector Unit

(48) The electromagnetic valve manifold **10** includes a connector member **70**. The connector

member **70** is disposed in the base **30**. The connector member **70** includes a base portion **71**. The connector member **70** includes a first base-side connector unit **80** and a second base-side connector unit **90**, which are base-side connector units. Therefore, the base **30** includes the first base-side connector unit **80** and the second base-side connector unit **90**, which are base-side connector units. The connector member **70** is formed by integrating the first base-side connector unit **80** and the second base-side connector unit **90** with the base portion **71**.

(49) As shown in FIG. 4, the base portion **71** has the shape of a flat plate. Specifically, the base portion **71** has the shape of a thin plate. The base portion **71** is made of plastic. The base portion **71** has the shape of an elongated rectangular plate in plan view in the thickness direction of the base portion **71**. The base portion **71** includes a through-hole **72**. The through-hole **72** extends through the base portion **71** in the thickness direction. The through-hole **72** is located in a center portion of the base portion **71**. In plan view in the thickness direction of the base portion **71**, the through-hole **72** has the shape of an elongated rectangular hole. The longitudinal direction of the through-hole **72** is the same as the longitudinal direction of the base portion **71**.

(50) As shown in FIG. 3, the first base-side connector unit **80** includes a first terminal accommodating portion **81** and first connection terminals **82**. The first terminal accommodating portion **81** is tubular and projects from the base portion **71**. The first terminal accommodating portion **81** includes two first terminal accommodating chambers **83**. The two first terminal accommodating chambers **83** are arranged in a direction orthogonal to the axial direction of the first terminal accommodating portion **81**. The two first terminal accommodating chambers **83** are spaced apart from each other by a first partition wall **84**, which is part of the first terminal accommodating portion **81**. Each first terminal accommodating chamber is open in a surface of the base portion **71** on a side opposite to the first terminal accommodating portion **81** through a corresponding first terminal insertion hole **85** formed in the base portion **71**. The first terminal accommodating portion **81** includes two first conductive member insertion holes **86** in a distal end face. Each first terminal accommodating chamber **83** opens in the distal end face of the first terminal accommodating portion **81** via the corresponding first conductive member insertion hole **86**.

(51) The first base-side connector unit **80** includes two first connection terminals **82**. The two first connection terminals **82** are respectively accommodated in the corresponding first terminal accommodating chambers **83**. Each first conductive member **52**, which is inserted into the corresponding first terminal accommodating chamber **83** through the corresponding first conductive member insertion hole **86**, can be inserted into and connected to one of the first connection terminals **82**. Thus, each first connection terminal **82** is accommodated in the corresponding first terminal accommodating chamber **83** such that the first conductive member **52**, which is inserted into the first terminal accommodating chamber **83** through the first conductive member insertion hole **86**, can be inserted into and connected to the first connection terminal **82**. In this manner, each first conductive member **52** is connected to the first base-side connector unit **80**.

(52) Also, the end of each first connection terminal **82** on the side opposite to the first conductive member **52** is electrically connected to the circuit board **41** via the corresponding first terminal insertion hole **85**. Thus, each first connection terminal **82** electrically connects the corresponding first conductive member **52** and the circuit board **41** to each other. Therefore, the first base-side connector unit **80**, to which the first electromagnetic valve-side connector unit **55** is connected, electrically connects the first electromagnetic valve-side connector unit **55** and the circuit board **41** to each other.

(53) The second base-side connector unit **90** includes a second terminal accommodating portion **91** and second connection terminals **92**. The second terminal accommodating portion **91** is tubular and projects from the base portion **71**. The second terminal accommodating portion **91** includes two second terminal accommodating chambers **93**. The two second terminal accommodating chambers **93** are arranged in a direction orthogonal to the axial direction of the second terminal

accommodating portion **91**. The two second terminal accommodating chambers **93** are spaced apart from each other by a second partition wall **94**, which is part of the second terminal accommodating portion **91**. Each second terminal accommodating chamber is open in a surface of the base portion **71** on a side opposite to the second terminal accommodating portion **91** through a corresponding second terminal insertion hole **95** formed in the base portion **71**. The second terminal accommodating portion **91** includes two second conductive member insertion holes **96** in a distal end face. Each second terminal accommodating chamber **93** opens in the distal end face of the second terminal accommodating portion **91** via the corresponding second conductive member insertion hole **96**.

(54) The second base-side connector unit **90** includes two second connection terminals **92**. The two second connection terminals **92** are respectively accommodated in the corresponding second terminal accommodating chambers **93**. Each second conductive member **62**, which is inserted into the corresponding second terminal accommodating chamber **93** through the corresponding second conductive member insertion hole **96**, can be inserted into and connected to one of the second connection terminals **92**. Thus, each second connection terminal **92** is accommodated in the corresponding second terminal accommodating chamber **93** such that the second conductive member **62**, which is inserted into the second terminal accommodating chamber **93** through the second conductive member insertion hole **96**, can be inserted into and connected to the second connection terminal **92**. In this manner, each second conductive member **62** is connected to the second base-side connector unit **90**.

(55) The end of each second connection terminal **92** on the side opposite to the second conductive member **62** is electrically connected to the circuit board **41** via the second terminal insertion hole **95**. Thus, each second connection terminal **92** electrically connects the corresponding second conductive member **62** and the circuit board **41** to each other. Therefore, the second base-side connector unit **90**, to which the second electromagnetic valve-side connector unit **65** is connected, electrically connects the second electromagnetic valve-side connector unit **65** and the circuit board **41** to each other.

(56) As shown in FIG. 2, the base **30** includes a fitting recess **47**. The fitting recess **47** is formed in the placement surface **31a**. The projecting portion **25** can be fitted in the fitting recess **47**. The fitting recess **47** includes a bottom surface **47a**, a first inner side surface **47b**, and a second inner side surface **47c**. The bottom surface **47a** is flat. The bottom surface **47a** extends parallel to the placement surface **31a**. The first inner side surface **47b** and the second inner side surface **47c** extend in a direction orthogonal to the bottom surface **47a**. The first inner side surface **47b** and the second inner side surface **47c** extend from the bottom surface **47a**. The first inner side surface **47b** and the second inner side surface **47c** are continuous with the placement surface **31a**. The first inner side surface **47b** and the second inner side surface **47c** connect the bottom surface **47a** and the placement surface **31a** to each other. The first inner side surface **47b** faces the first side surface **25a** of the projecting portion **25**. The second inner side surface **47c** faces the second side surface **25b** of the projecting portion **25**.

(57) As shown in FIG. 3, the base **30** includes an attachment wall portion **42**. The connector member **70** is attached to the attachment wall portion **42**. The outer surface of the attachment wall portion **42** is the bottom surface **47a** of the fitting recess **47**. The inner surface of the attachment wall portion **42** is an attachment surface **42b** on a side opposite to the bottom surface **47a** of the fitting recess **47**. The connector member **70** is attached to the attachment surface **42b**. The attachment surface **42b** is flat.

(58) The base **30** includes a first insertion port **43**, which is an insertion port. The first insertion port **43** extends through the attachment wall portion **42**. The first insertion port **43** opens in the bottom surface **47a** of the fitting recess **47**. The first terminal accommodating portion **81** is disposed inside the first insertion port **43**. Therefore, the first base-side connector unit **80** is disposed inside the first insertion port **43**. The first tubular portion **53** is inserted into the first insertion port **43**. Therefore,

the first electromagnetic valve-side connector unit **55** is inserted into the first insertion port **43**. The first terminal accommodating portion **81** is inserted into the first tubular portion **53**. Therefore, the first base-side connector unit **80** is inserted into the first tubular portion **53**. The first lip packing **54** is provided between the outer circumferential surface of the first tubular portion **53** and the inner circumferential surface of the first insertion port **43**. The first lip packing **54** provides a seal between the first tubular portion **53** and the first insertion port **43**.

(59) The base **30** includes a second insertion port **44**, which is an insertion port. The second insertion port **44** extends through the attachment wall portion **42**. The second insertion port **44** opens in the bottom surface **47a** of the fitting recess **47**. The first insertion port **43** and the second insertion port **44** are arranged side by side in the arrangement direction of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**. Therefore, the arrangement direction of the first insertion port **43** and the second insertion port **44** is the same as the arrangement direction of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**.

(60) The second terminal accommodating portion **91** is disposed inside the second insertion port **44**. Therefore, the second base-side connector unit **90** is disposed inside the second insertion port **44**. The second tubular portion **63** is inserted into the second insertion port **44**. Therefore, the second electromagnetic valve-side connector unit **65** is inserted into the second insertion port **44**. The second terminal accommodating portion **91** is inserted into the second tubular portion **63**. Therefore, the second base-side connector unit **90** is inserted into the second tubular portion **63**. The second lip packing **64** is provided between the outer circumferential surface of the second tubular portion **63** and the inner circumferential surface of the second insertion port **44**. The second lip packing **64** provides a seal between the second tubular portion **63** and the second insertion port **44**.

(61) The base **30** includes a projecting wall **45**. The projecting wall **45** is columnar. The projecting wall **45** projects from the attachment wall portion **42**. In plan view in the thickness direction of the base portion **71**, the projecting wall **45** has the shape of an elongated rectangle as shown in FIG. 4. The base **30** includes two engagement projections **46**. The two engagement projections **46** respectively project from the projecting wall **45**. A section of the base portion **71** around the through-hole **72** can be engaged with the two engagement projections **46**.

(62) The base portion **71** is elastically deformable when the two engagement projections **46** are inserted into the through-hole **72**. Specifically, when the two engagement projections **46** are inserted into the through-hole **72**, the section of the base portion **71** around the through-hole **72** is easily elastically deformed in the traverse direction of the base portion **71**. When the two engagement projections **46** pass through the through-hole **72**, the section of the base portion **71** around the through-hole **72** returns to its original shape before the elastic deformation. The section of the base portion **71** around the through-hole **72** is engaged with the two engagement projections **46**.

(63) In this manner, the connector member **70** is attached to the base **30** by engaging the section of the base portion **71** around the through-hole **72** with the two engagement projections **46**. In a state in which the connector member **70** is attached to the base **30**, the connector member **70** is allowed to move slightly in the longitudinal direction and the traverse direction of the base portion **71**.

(64) As shown in FIG. 3, when the first tubular portion **53** is inserted into the first insertion port **43**, each first conductive member **52** is inserted into the first terminal accommodating chamber **83** via the first conductive member insertion hole **86** to be inserted into and connected to the corresponding first connection terminal **82**. In this manner, the first electromagnetic valve-side connector unit **55** is connected to the first base-side connector unit **80**. Also, when the second tubular portion **63** is inserted into the second insertion port **44**, each second conductive member **62** is inserted into the second terminal accommodating chamber **93** via the second conductive member insertion hole **96** to be inserted into and connected to the corresponding second connection terminal

92. In this manner, the second electromagnetic valve-side connector unit **65** is connected to the second base-side connector unit **90**.

(65) The insertion direction of the first electromagnetic valve-side connector unit **55** into the first insertion port **43** is the same as the insertion direction of the second electromagnetic valve-side connector unit **65** into the second insertion port **44**. In the following description, the insertion direction of the first electromagnetic valve-side connector unit **55** into the first insertion port **43** and the insertion direction of the second electromagnetic valve-side connector unit **65** into the second insertion port **44** will be simply referred to as an insertion direction **Z1** in some cases.

(66) Protrusions and Recesses

(67) As shown in FIG. 5, the electromagnetic valve **11** includes a first protrusion **56** and a second protrusion **66**, which are protrusions. The first protrusion **56** and the second protrusion **66** are provided on the projecting portion **25**. The first protrusion **56** protrudes from the first side surfaces **25a** of the projecting portion **25**. The second protrusion **66** protrudes from the second side surface **25b** of the projecting portion **25**. The first protrusion **56** and the second protrusion **66** each have the shape of a quadrangular prism and extend in the insertion direction **Z1**. An end face of each of the first protrusion **56** and the second protrusion **66** that faces the base **30** protrudes further than the distal end face of the projecting portion **25**.

(68) As shown in FIGS. 5 and 6, the base **30** includes a first recess **87** and a second recess **97**, which are recesses. The first recess **87** is formed in the first inner side surface **47b**. The first recess **87** opens in the first inner side surface **47b** and the placement surface **31a**. The first protrusion **56** is inserted into the first recess **87**. The insertion direction of the first protrusion **56** into the first recess **87** is the same as the insertion direction **Z1**. The first recess **87** guides the first protrusion **56** in the same direction as the insertion direction **Z1**. The first recess **87** includes a first open end, which opens in the placement surface **31a**, and a first closed end, which is located on a side opposite to the first open end, in the insertion direction of the first protrusion **56** into the first recess **87**. The first open end is a receiving port that receives the first protrusion **56**.

(69) The second recess **97** is formed in the second inner side surface **47c**. The second recess **97** opens in the second inner side surface **47c** and the placement surface **31a**. The second protrusion **66** is inserted into the second recess **97**. The insertion direction of the second protrusion **66** into the second recess **97** is the same as the insertion direction **Z1**. The second recess **97** guides the second protrusion **66** in the same direction as the insertion direction **Z1**. The second recess **97** includes a second open end, which opens in the placement surface **31a**, and a second closed end, which is located on a side opposite to the second open end, in the insertion direction of the second protrusion **66** into the second recess **97**. The second open end is a receiving port that receives the second protrusion **66**.

(70) As shown in FIG. 7, the second recess **97** includes a bottom surface **97a**, first straight surfaces **97b**, first tapered surfaces **97c**, second straight surfaces **97d**, and second tapered surfaces **97e**. The bottom surface **97a** is a surface of the second recess **97** that is located at the innermost position in the insertion direction of the second protrusion **66** into the second recess **97**. The bottom surface **97a** is located at the second closed end of the second recess **97**. The bottom surface **97a** is continuous with the bottom surface **47a** of the fitting recess **47**. The bottom surface **97a** is located on the same plane as the bottom surface **47a** of the fitting recess **47**.

(71) The first straight surfaces **97b** are continuous with the bottom surface **97a**. The first straight surfaces **97b** extend in a direction orthogonal to the bottom surface **97a**. The first straight surfaces **97b** extend along side surfaces of the second protrusion **66**. The first tapered surfaces **97c**, which are first inclined surfaces, are continuous with edges of the first straight surfaces **97b** on a side opposite to the bottom surface **97a**. The first tapered surfaces **97c** are inclined such that the width of the second recess **97** is larger at any position further from the first straight surfaces **97b** than at a position closer to the first straight surfaces **97b**. The width of the second recess **97** is the dimension in the left-right direction as viewed in FIG. 7. The second straight surfaces **97d** are continuous with

edges of the first tapered surfaces **97c** on a side opposite to the first straight surfaces **97b**. The second straight surfaces **97d** extend parallel to the first straight surfaces **97b**.

(72) The second tapered surfaces **97e**, which are second inclined surfaces, are continuous with edges of the second straight surfaces **97d** on a side opposite to the first tapered surfaces **97c**. The second tapered surfaces **97e** are inclined such that the width of the second recess **97** is larger at any position further from the second straight surfaces **97d** than at a position closer to the second straight surfaces **97d**. The second tapered surfaces **97e** are continuous with the placement surface **31a**. The second tapered surfaces **97e** are surfaces of the second recess **97** that are closest to the receiving port in the insertion direction of the second protrusion **66** into the second recess **97**. The second recess **97** includes a receiving port, which receives the second protrusion **66**, and a portion other than the receiving port. The portion other than the receiving port is an inner section that is located further inward than the receiving port in the insertion direction of the second protrusion **66** into the second recess **97**. The second recess **97** has a larger width at the receiving port than at the inner section. The second recess **97** includes a wide portion and a narrow portion. The wide portion includes the receiving port (the second open end). The narrow portion is a section other than the wide portion and has a smaller width than that of the wide portion. For example, the wide portion corresponds to the second tapered surfaces **97e**, and the narrow portion corresponds to the section other than the second tapered surfaces **97e**.

(73) The relationship between the first protrusion **56** and the first recess **87** is substantially the same as the relationship between the second protrusion **66** and the second recess **97**. Therefore, as in the case of the second recess **97**, the first recess **87** has a width larger at the receiving port than at other sections. The first recess **87** includes a wide portion and a narrow portion. The wide portion includes the receiving port (the first open end). The narrow portion is a section other than the wide portion and has a smaller width than that of the wide portion. In FIG. 7, the same reference numerals are given to common components, and detailed explanation of the relationship between the first protrusion **56** and the first recess **87** will be omitted. For example, the first recess **87** of the present embodiment does not include the first tapered surfaces **97c** or the second straight surfaces **97d** as shown in FIG. 5, but may include the first tapered surfaces **97c** and the second straight surfaces **97d**.

(74) As shown in FIGS. 8 and 9, when the electromagnetic valve **11** is mounted to the base **30**, the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**. When the first protrusion **56** is inserted into the receiving port of the first recess **87** and the second protrusion **66** is inserted into the receiving port of the second recess **97**, the first tubular portion **53** has not been inserted into the first insertion port **43**, and the second tubular portion **63** has not been inserted into the second insertion port **44**. Therefore, the first protrusion **56** and the second protrusion **66** are designed to be inserted into the first recess **87** and the second recess **97** before the first tubular portion **53** and the second tubular portion **63** are inserted into the first insertion port **43** and the second insertion port **44**, respectively. Then, when the first protrusion **56** is guided to a section further inward than the receiving port in the first recess **87** and the second protrusion **66** is guided to a section further inward than the receiving port in the second recess **97**, the first tubular portion **53** is inserted into the first insertion port **43**, and the second tubular portion **63** is inserted into the second insertion port **44**. For example, when the distal end portions of the first and second protrusions **56**, **66** in the insertion direction pass through the wide portions (the second tapered surfaces **97e**) of the first and second recesses **87**, **97**, the first and second tubular portions **53**, **63** start to be inserted into the first and second insertion ports **43**, **44**.

(75) For example, as shown in FIG. 9, the first protrusion **56** is not in contact with the first recess **87** in a state in which the second protrusion **66** is in contact with the second recess **97**. At this time, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with a section of the base **30** around the first insertion port **43**, and the second tubular portion **63** is inserted into the second insertion port **44** without interfering with a section of the base **30** around

the second insertion port **44**. Also, the second protrusion **66** is not in contact with the second recess **97** in a state in which the first protrusion **56** is in contact with the first recess **87**. At this time also, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with a section of the base **30** around the first insertion port **43**, and the second tubular portion **63** is inserted into the second insertion port **44** without interfering with a section of the base **30** around the second insertion port **44**.

(76) As the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the first terminal accommodating portion **81**. Then, each first conductive member **52** is inserted into the corresponding first terminal accommodating chamber **83** via the first conductive member insertion hole **86** to be inserted into and connected to the corresponding first connection terminal **82**. In this manner, when the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the first base-side connector unit **80**. Each first conductive member **52** is thus connected to the first base-side connector unit **80**.

(77) As the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the second terminal accommodating portion **91**. Then, each second conductive member **62** is inserted into the corresponding second terminal accommodating chamber **93** via the second conductive member insertion hole **96** to be inserted into and connected to the corresponding second connection terminal **92**. In this manner, when the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the second base-side connector unit **90**. Each second conductive member **62** is thus connected to the second base-side connector unit **90**.

(78) Therefore, when the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, the first electromagnetic valve-side connector unit **55** is inserted into the first insertion port **43** without interfering with the section of the base **30** around the first insertion port **43**. The first electromagnetic valve-side connector unit **55** is thus connected to the first base-side connector unit **80**. Also, when the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, the second electromagnetic valve-side connector unit **65** is inserted into the second insertion port **44** without interfering with the section of the base **30** around the second insertion port **44**. The second electromagnetic valve-side connector unit **65** is thus connected to the second base-side connector unit **90**.

(79) When the first protrusion **56** is guided to a section further inward than the receiving port in the first recess **87** and the second protrusion **66** is guided to a section further inward than the receiving port in the second recess **97**, the first electromagnetic valve-side connector unit **55** is positioned in a location where the first tubular portion **53** does not interfere with the first base-side connector unit **80**. When the first protrusion **56** is guided to a section further inward than the receiving port in the first recess **87** and the second protrusion **66** is guided to a section further inward than the receiving port in the second recess **97**, the second electromagnetic valve-side connector unit **65** is positioned in a location where the second tubular portion **63** does not interfere with the second base-side connector unit **90**.

(80) Also, when the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, relative positions of each bolt insertion hole **38h** and the corresponding internal thread hole **39** agree with each other.

Operation of Embodiment

(81) Operation of the present embodiment will now be described.

(82) When connecting the first electromagnetic valve-side connector unit **55** to the first base-side

connector unit **80** and connecting the second electromagnetic valve-side connector unit **65** to the second base-side connector unit **90**, the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**. Accordingly, the first electromagnetic valve-side connector unit **55** is inserted into the first insertion port **43** without interfering with the section of the base **30** around the first insertion port **43**. Specifically, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the section of the base **30** around the first insertion port **43**. Further, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the first base-side connector unit **80**, so that each first conductive member **52** is connected to the first base-side connector unit **80**.

(83) Also, the second electromagnetic valve-side connector unit **65** is inserted into the second insertion port **44** without interfering with the section of the base **30** around the second insertion port **44**. Specifically, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the section of the base **30** around the second insertion port **44**. Further, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the second base-side connector unit **90**, so that each second conductive member **62** is connected to the second base-side connector unit **90**. Accordingly, the first electromagnetic valve-side connector unit **55** is smoothly connected to the first base-side connector unit **80**, and the second electromagnetic valve-side connector unit **65** is smoothly connected to the second base-side connector unit **90**.

(84) When each first conductive member **52** is connected to the first base-side connector unit **80** and each second conductive member **62** is connected to the second base-side connector unit **90**, the connector member **70** is allowed to move slightly in the longitudinal direction and the traverse direction of the base portion **71**. This facilitates the connection of each first conductive member **52** to the first base-side connector unit **80** and the connection of each second conductive member **62** to the second base-side connector unit **90**.

Advantages of Embodiment

(85) The above-described embodiment has the following advantages.

(86) (1) The electromagnetic valve **11** includes the first protrusion **56** and the second protrusion **66**. The base **30** includes the first recess **87**, which guides the first protrusion **56**, and the second recess **97**, which guides the second protrusion **66**. When the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, the first electromagnetic valve-side connector unit **55** is inserted into the first insertion port **43** without interfering with the section of the base **30** around the first insertion port **43**. The first electromagnetic valve-side connector unit **55** is thus connected to the first base-side connector unit **80**. Also, when the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, the second electromagnetic valve-side connector unit **65** is inserted into the second insertion port **44** without interfering with the section of the base **30** around the second insertion port **44**. The second electromagnetic valve-side connector unit **65** is thus connected to the second base-side connector unit **90**. This allows the first electromagnetic valve-side connector unit **55** to be smoothly connected to the first base-side connector unit **80**, and the second electromagnetic valve-side connector unit **65** to be smoothly connected to the second base-side connector unit **90**. This improves the ease of mounting for the electromagnetic valve manifold **10**.

(87) (2) When the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the first base-side connector unit **80**. Each first conductive member **52** is thus connected to the first base-side connector unit **80**. When the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the second base-side connector unit **90**. Each second conductive member **62** is thus connected to the second base-side connector unit **90**. This allows the first electromagnetic valve-side connector unit **55** to be

smoothly connected to the first base-side connector unit **80**, and the second electromagnetic valve-side connector unit **65** to be smoothly connected to the second base-side connector unit **90**. This further improves the ease of mounting for the electromagnetic valve manifold **10**.

(88) (3) The width of the first recess **87** is larger at the receiving port, which receives the first protrusion **56**, than at the inner section, which is located further inward than the receiving port. This allows the first protrusion **56** to be easily inserted into the first recess **87**. The width of the second recess **97** is larger at the receiving port, which receives the second protrusion **66**, than at the inner section, which is located further inward than the receiving port. This allows the second protrusion **66** to be easily inserted into the second recess **97**. When the first protrusion **56** is guided to a section further inward than the receiving port in the first recess **87** and the second protrusion **66** is guided to a section further inward than the receiving port in the second recess **97**, the first electromagnetic valve-side connector unit **55** is positioned in a location where the first tubular portion **53** does not interfere with the first base-side connector unit **80**. When the first protrusion **56** is guided to a section further inward than the receiving port in the first recess **87** and the second protrusion **66** is guided to a section further inward than the receiving port in the second recess **97**, the second electromagnetic valve-side connector unit **65** is positioned in a location where the second tubular portion **63** does not interfere with the second base-side connector unit **90**. Therefore, even if the width of the receiving port of each of the first recess **87** and the second recess **97** is relatively large, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the first base-side connector unit **80**. Each first conductive member **52** is thus connected to the first base-side connector unit **80**. Also, even if the width of the receiving port of each of the first recess **87** and the second recess **97** is relatively large, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the second base-side connector unit **90**. Each second conductive member **62** is thus connected to the second base-side connector unit **90**. This further improves the ease of mounting for the electromagnetic valve manifold **10**.

(89) (4) When the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**, relative positions of each bolt insertion hole **38h** and the corresponding internal thread hole **39** agree with each other. This configuration prevents the bolts **38** from interfering with the gasket **37** when the bolts **38** are inserted into the bolt insertion holes **38h** and threaded into the respective internal thread holes **39**.

Modifications

(90) The above-described embodiment may be modified as follows. The above-described embodiment and the following modifications can be combined if the combined modifications remain technically consistent with each other.

(91) As shown in FIG. **10**, the second recess **97** may include holding portions **97f**. The holding portions **97f** hold the second protrusion **66**. The holding portions **97f**, for example, are projections that project from two opposed surfaces on the first straight surfaces **97b** of the second recess **97**. The holding portions **97f** hold the second protrusion **66**, which is inserted into the second recess **97**, in a clamping manner.

(92) With this configuration, the second protrusion **66** is held by the holding portions **97f**. This prevents the electromagnetic valve **11** from separating from the base **30** after the first electromagnetic valve-side connector unit **55** is connected to the first base-side connector unit **80** and the second electromagnetic valve-side connector unit **65** is connected to the second base-side connector unit **90**. The first recess **87** may also include holding portions **97f**. The holding portions **97f** of the first recess **87** hold the first protrusion **56**.

(93) As shown in FIG. **11**, the second recess **97** does not necessarily need to have a larger width at the receiving port than at the inner section. A distal end portion **66a** of the second protrusion **66** in the insertion direction into the second recess **97** may be thinner than sections other than the distal end portion **66a**.

(94) This allows the second protrusion **66** to be easily inserted into the second recess **97**. When the

section of the second protrusion **66** other than the distal end portion **66a** is guided by the second recess **97**, the first electromagnetic valve-side connector unit **55** is positioned in a location where the first tubular portion **53** does not interfere with the first base-side connector unit **80**. When the section of the second protrusion **66** other than the distal end portion **66a** is guided by the second recess **97**, the second electromagnetic valve-side connector unit **65** is positioned in a location where the second tubular portion **63** does not interfere with the second base-side connector unit **90**.

Therefore, even if the distal end portion **66a** in the insertion direction of the second protrusion **66** into the second recess **97** is thinner than the sections other than the distal end portion **66a**, the first tubular portion **53** is inserted into the first insertion port **43** without interfering with the first base-side connector unit **80**. Each first conductive member **52** is thus connected to the first base-side connector unit **80**. Also, even if the distal end portion **66a** in the insertion direction of the second protrusion **66** into the second recess **97** is thinner than the sections other than the distal end portion **66a**, the second tubular portion **63** is inserted into the second insertion port **44** without interfering with the second base-side connector unit **90**. Each second conductive member **62** is thus connected to the second base-side connector unit **90**. This further improves the ease of mounting for the electromagnetic valve manifold **10**.

(95) The first recess **87** also does not necessarily need to have a larger width at the receiving port than at the inner section. A distal end portion of the first protrusion **56** in the insertion direction into the first recess **87** may also be thinner than sections other than the distal end portion.

(96) In the above-described embodiment, for example, the base **30** may include a protrusion, and the electromagnetic valve **11** may include a recess that guides the protrusion in the same direction as the insertion direction **Z1**. That is, it suffices if one of the electromagnetic valve **11** and the base **30** includes a protrusion, and the other one of the electromagnetic valve **11** and the base **30** includes a recess.

(97) In the above-described embodiment, the first electromagnetic valve-side connector unit **55** does not necessarily need to include the first tubular portion **53**. Also, the second electromagnetic valve-side connector unit **65** does not necessarily need to include the second tubular portion **63**. In this case also, it suffices if each first conductive member **52** is inserted into the first insertion port **43** without interfering with the section of the base **30** around the first insertion port **43** as the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**. Also, it suffices if each first conductive member **52** is connected to the first base-side connector unit **80**. Further, it suffices if each second conductive member **62** is inserted into the second insertion port **44** without interfering with the section of the base **30** around the second insertion port **44** as the first protrusion **56** is guided by the first recess **87** and the second protrusion **66** is guided by the second recess **97**. Also, it suffices if each second conductive member **62** is connected to the second base-side connector unit **90**.

(98) In the above-described embodiment, the first recess **87** may include, in addition to the bottom surface **97a**, the first straight surfaces **97b**, and the second tapered surfaces **97e**, first tapered surfaces **97c** and second straight surfaces **97d** as in the case of the second recess **97**.

(99) In the above-described embodiment, the first tapered surfaces **97c** and the second straight surfaces **97d** may be omitted from the second recess **97**.

(100) In the above-described embodiment, the first recess **87** does not necessarily need to have a larger width at the receiving port than at the inner section. Also, the second recess **97** does not necessarily need to have a larger width at the receiving port than at the inner section.

(101) In the above-described embodiment, when the first protrusion **56** is guided by the first recess **87**, and the second protrusion **66** is guided by the second recess **97**, relative positions of each bolt insertion hole **38h** and the corresponding internal thread hole **39** do not necessarily need to agree with each other.

(102) In the above-described embodiment, an annular gasket may be provided between the base **30** and the set of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**.

The gasket may provide a seal between the base **30** and the set of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**. In this case, the electromagnetic valve manifold **10** does not necessarily need to include the first lip packing **54** and the second lip packing **64**.

(103) Even if a gasket is provided between the base **30** and the set of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**, the first electromagnetic valve-side connector unit **55** is inserted into the first insertion port **43** without interfering with the gasket. Also, the second electromagnetic valve-side connector unit **65** is inserted into the second insertion port **44** without interfering with the gasket. Therefore, it is possible to avoid a sealing failure between the base **30** and the set of the first pilot electromagnetic valve **50** and the second pilot electromagnetic valve **60**.

(104) In the above-described embodiment, for example, the electromagnetic valve **11** may include only one protrusion, and the base **30** may include only one recess that guides the protrusion in the same direction as the insertion direction **Z1**. In this case, when the protrusion has the form of a quadrangular prism, the recess formed in the placement surface **31a** is preferably, for example, a quadrangular hole-shaped recess. Any modification may be made as long as the protrusion is guided by the recess such that the first electromagnetic valve-side connector unit **55** is inserted into the first insertion port **43** and connected to the first base-side connector unit **80** without interfering with the section of the base **30** around the first insertion port **43**. Also, any modification may be made as long as the protrusion is guided by the recess such that the second electromagnetic valve-side connector unit **65** is inserted into the second insertion port **44** and connected to the second base-side connector unit **90** without interfering with the section of the base **30** around the second insertion port **44**.

(105) In the above-described embodiment, the number of the first conductive members **52** is not particularly limited.

(106) In the above-described embodiment, the number of the second conductive members **62** is not particularly limited.

(107) In the above-described embodiment, the electromagnetic valve **11** includes the first electromagnetic valve-side connector unit **55** and the second electromagnetic valve-side connector unit **65**. Instead, for example, the electromagnetic valve **11** may include only one electromagnetic valve-side connector unit. In this case, the electromagnetic valve-side connector unit includes a single tubular portion, a conductive member that is electrically connected to the first solenoid unit **51**, and a conductive member that is electrically connected to the second solenoid unit **61**. The single tubular portion surrounds both conductive members. The base **30** includes a single base-side connector unit.

(108) In the above-described embodiment, the electromagnetic valve **11** may be a single-solenoid electromagnetic pilot valve. In this case, the electromagnetic valve **11** includes only one electromagnetic valve-side connector unit. The base **30** includes a single base-side connector unit.

(109) In the above-described embodiment, the electromagnetic valve **11** may be a four-port electromagnetic valve from which, for example, the second discharge port **R2** is omitted. That is, any type of electromagnetic valve may be used as the electromagnetic valve **11** as long as the electromagnetic valve **11** includes at least one discharge port. Also, the electromagnetic valve **11** may be a three-port electromagnetic valve that includes a supply port, an output port, and a discharge port.

(110) The term “annular” as used in this description may refer to any structure that forms a loop as a whole. “Annular” shapes include but are not limited to a circular shape, an elliptic shape, and a polygonal shape with sharp or rounded corners. “Tubular” shapes include but are not limited to any structure having circular cross-sectional shape, an elliptic cross-sectional shape, and a polygonal cross-sectional shape with sharp or rounded corners.

Claims

1. An electromagnetic valve manifold, comprising: an electromagnetic valve including a solenoid unit; a control unit including a circuit board configured to control operation of the electromagnetic valve; and a base that includes the control unit, wherein the electromagnetic valve includes an electromagnetic valve-side connector unit electrically connected to the solenoid unit, the base includes a base-side connector unit to which the electromagnetic valve-side connector unit is connected, the base-side connector unit is configured to electrically connect the electromagnetic valve-side connector unit and the circuit board to each other, the electromagnetic valve-side connector unit includes: a conductive member electrically connected to the solenoid unit; and a tubular portion that surrounds the conductive member, the base-side connector unit being inserted into the tubular portion, the base includes an insertion port into which the electromagnetic valve-side connector unit is inserted, the base-side connector unit is disposed inside the insertion port, one of the electromagnetic valve or the base includes a protrusion, the other one of the electromagnetic valve or the base includes a recess, the recess being configured to guide the protrusion in a same direction as an insertion direction of the electromagnetic valve-side connector unit into the insertion port, and when the protrusion is guided by the recess, the electromagnetic valve-side connector unit is inserted into the insertion port without interfering with a section of the base around the insertion port and is connected to the base-side connector unit, and the tubular portion is inserted into the insertion port without interfering with the base-side connector unit, and the conductive member is connected to the base-side connector unit.
 2. The electromagnetic valve manifold according to claim 1, wherein the recess includes: a receiving port that receives the protrusion; and an inner section that is located further inward than the receiving port in an insertion direction of the protrusion into to the recess, the recess has a larger width at the receiving port than at the inner section, and the electromagnetic valve-side connector unit is configured such that, when the protrusion is guided to the inner section of the recess, the tubular portion is positioned in a location where the tubular portion does not interfere with the base-side connector unit.
 3. The electromagnetic valve manifold according to claim 1, wherein the protrusion includes: a distal end portion in an insertion direction of the protrusion into to the recess; and a portion other than the distal end portion, the distal end portion is thinner than the portion other than the distal end portion, and the electromagnetic valve-side connector unit is configured such that, when the portion other than the distal end portion is guided by the recess, the tubular portion is positioned in a location where the tubular portion does not interfere with the base-side connector unit.
 4. The electromagnetic valve manifold according to claim 1, wherein the electromagnetic valve is fixed to the base with a bolt, the electromagnetic valve includes a bolt insertion hole through which the bolt is inserted, the base includes an internal thread hole into which the bolt is threaded, a gasket is arranged between the electromagnetic valve and the base, and as the protrusion is guided by the recess, relative positions of the bolt insertion hole and the internal thread hole agree with each other.
 5. The electromagnetic valve manifold according to claim 1, wherein the recess includes a holding portion configured to hold the protrusion.
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